Research scenario

HEOR technical dossier

Every modeled value, its basis and its evidence: the public edition

PRIMARY MODELED NET MONETARY BENEFIT ASSUMPTION-DRIVEN SCENARIO

$40,160

Lynch syndrome carrier identified

Model basis: a published model's printed outputs, restated by stated arithmetic where the source prints another unit.

4
modeled values, each with a model card

12
of 28 CHEERS 2022 items addressed (8 in part)

3
checks against sources; none is external validation

The main caveat

- Research-genome calls, not confirmed by a clinical laboratory: not diagnostic results. The age 30 is stated for this run, not observed: it is the age at which the published source reports its result, so the donor-specific economic value is not established.

Inside

1. **Objective and cost-effectiveness plane** — each decision against its own comparator
2. **Action versus inaction** — the lead value's health gain, cost and net monetary benefit
3. **Model cards** — decision, comparator, population, perspective, horizon, assumptions and sources
4. **CHEERS 2022 map** — where each reporting item is addressed, and which are not
5. **Validation record and limitations** — what was checked against published sources, and what was not

Proprietary. Demonstration report for a public research genome. GenomeLens, its economic models and its methods are proprietary. This report shows results, not the method: it is not open-source software, not a description sufficient to reproduce the analysis, and not a license to use either. This edition shows each value, its comparator, population, perspective, horizon, assumptions and sources; it does not publish the model's inputs or structure.

Every modeled value, its basis and its evidence

Objective: to value, for one genome, each clinical decision its findings change, as net monetary benefit at $100,000 per QALY against the decision's own comparator, from published cost-effectiveness evidence, under a standardized age, with every assumption named. Each modeled value below has a model card: its decision and comparator, population, perspective, horizon, discounting, price year, key assumptions and sources. A card states a value's basis (a published model's printed outputs, restated, or a hypothetical scenario built here from published components) where one is labeled. The model's inputs and structure are not published. The arithmetic that restates the published source's printed figures as a value per carrier is set out in docs/ECONOMICS.md (How the age-30 value is derived).

Where each modeled decision falls

FIGURE 1



primary modeled cascade testing (relatives) alternative model

- - - $100,000 per QALY: below the line and right of the axis is cost-effective at that threshold

	TABLE 1. THE PLOTTED VALUES	ΔQALY	ΔCOST	NMB
1	**Primary modeled**: Lynch syndrome carrier identified Colonoscopy surveillance after identification · ASSUMPTION-DRIVEN SCENARIO	+0.5091	+$10,754	$40,160
2	**Cascade testing (relatives)**: Lynch syndrome carrier identified Colonoscopy surveillance after identification · DIFFERENT BENEFICIARY · ASSUMPTION-DRIVEN SCENARIO	+0.11	+$3,795	$7,205
3	**Alternative model**: Lynch syndrome carrier identified (Bennette 2015) Return of the finding, then guideline management · ASSUMPTION-DRIVEN SCENARIO	+0.95	+$4,508	$90,492
4	**Alternative model**: MSH2 carrier identified (man) Colonoscopy every 2 years to 75 · ASSUMPTION-DRIVEN SCENARIO	+0.3828	+$5,514	$32,766

Each point is a different model or beneficiary (own comparator, population, horizon); no two are added.

Basis of the lead value. Model basis: a published model's printed outputs, restated by stated arithmetic where the source prints another unit.

11
economic results assessed

4
modeled values, each with a model card

12
of 28 CHEERS 2022 items addressed (8 in part)

3
checks against sources; none is external validation

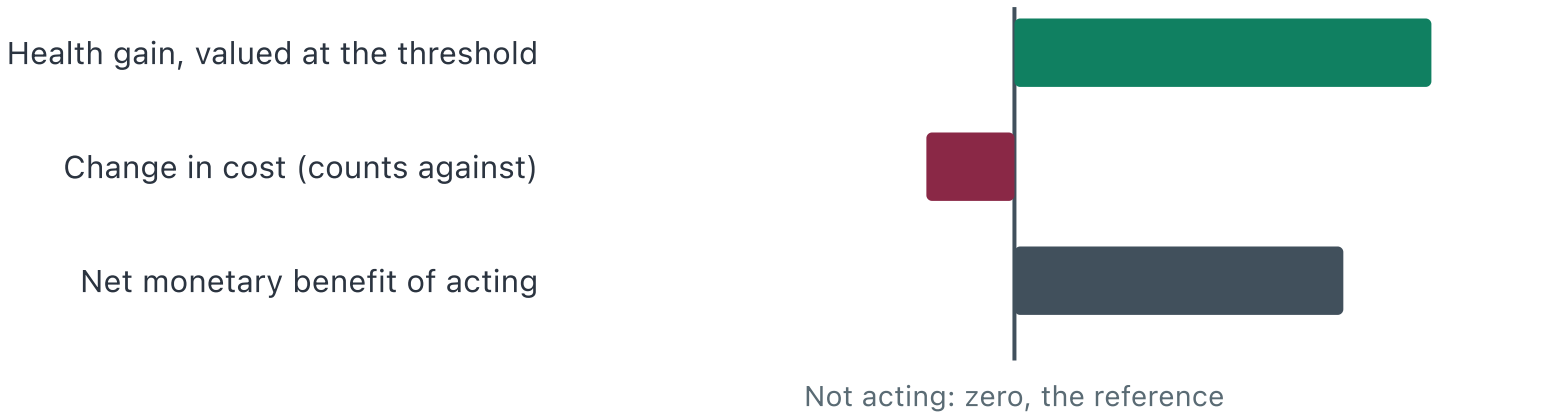
State: the figures on this page are model outputs, labeled UNRESOLVED (not established for this donor) unless a row carries another label. Standardized scenario at assumed age 30: the donor-specific economic value is not established with available context.

Research scenario

Acting on the lead finding against not acting

Acting on the finding against not acting

FIGURE 2



TERM, ACTING AGAINST NOT ACTING	PER PERSON
Health gain	**+0.5091 QALYs**
Change in cost	**+$10,754**
Net monetary benefit at $100,000 per QALY	**$40,160** ASSUMPTION-DRIVEN SCENARIO

Lynch syndrome carrier identified: Colonoscopy surveillance after identification, against family history-based testing alone (the status quo). End totals only: a transferred published result has no per-year course, and none is drawn.

Model basis: a published model's printed outputs, restated by stated arithmetic where the source prints another unit.

Model cards

Research scenario

HEOR TECHNICAL DOSSIER · HU4AE0F2

1 Primary modeled: Lynch syndrome carrier identified

$40,160

ΔQALY
+0.5091 QALYs

ΔCOST
+$10,754

NET MONETARY BENEFIT
$40,160
ASSUMPTION-DRIVEN SCENARIO

5TH TO 95TH PERCENTILE
$25,045 to $56,294

COST PER QALY GAINED
$21,121 per QALY gained, below the $100,000 threshold

DECISION
Colonoscopy surveillance after identification

COMPARATOR
Family history-based testing alone (the status quo)

PERSPECTIVE
US, health-care sector

POPULATION
US adults in population genomic screening (four Lynch genes pooled; applied to MLH1 and MSH2)

HORIZON
Lifetime

DISCOUNTING
3% a year

AGE
At the stated age of 30, assumed for this run, not observed; if identified at 30

PRICE YEAR
Source 2021, restated to 2019 USD

VALUED PER
Carrier in the source population (per person screened, divided by carrier prevalence; the source's 200 USD test excluded as sunk)

Model basis: a published model's printed outputs, restated by stated arithmetic where the source prints another unit.

Genome-first screening value, a different question (not netted, not added): a Lynch-only test at 200 USD per person is not cost-effective in Guzauskas 2022 (ICER 153,200 USD per QALY without cascade testing), so its net monetary benefit is negative per person screened and per carrier found; the value above starts after sequencing, so the test cost is sunk.

Key assumptions. At the stated age of 30, assumed for this run, not observed; age stated, not observed.

Secondary comparison. Kastrinos 2021, PMID 33839100, Table 2 compares colonoscopy schedules among carriers who are already known, identified at 25, with perfect adherence, in 2019 US dollars. It has no 'not identified' arm, so it is not this decision's value and is in no total. The source's optimal schedule for MSH2 (every 2 years from age 25) against the next efficient schedule (every 3 years from age 25): +0.028 QALYs, +908 USD, a net monetary benefit of 1,892 USD at 100,000 USD per QALY. Against the least intensive schedule it lists (every 5 years from age 40, current management for a family history, which the source rates dominated): +0.664 QALYs, −25,131 USD, a net monetary benefit of 91,531 USD. One schedule against two comparators: the figures are not interchangeable. Its discounting is unresolved: the printed QALYs exceed what its stated 3% can give, so these figures are arithmetic on the totals as printed.

Sources. PMID 35227606.

2 Cascade testing (relatives): Lynch syndrome carrier identified

$7,205

ΔQALY
+0.11 QALYs

ΔCOST
+$3,795

NET MONETARY BENEFIT
$7,205
DIFFERENT BENEFICIARY · ASSUMPTION-DRIVEN SCENARIO

UNCERTAINTY
not sampled (an alternative or cascade row)

DECISION
Colonoscopy surveillance after identification

COMPARATOR
Family history-based testing alone (the status quo)

PERSPECTIVE
US, health-care sector

POPULATION
US adults in population genomic screening (four Lynch genes pooled; applied to MLH1 and MSH2)

HORIZON
Lifetime

DISCOUNTING
3% a year

AGE
At the stated age of 30, assumed for this run, not observed

PRICE YEAR
Source 2021, restated to 2019 USD

VALUED PER
Carrier in the source population (per person screened, divided by carrier prevalence; the source's 200 USD test excluded as sunk)

Key assumptions. At the stated age of 30, assumed for this run, not observed; age stated, not observed.

Sources. PMID 35227606.

3 Alternative model: Lynch syndrome carrier identified (Bennette 2015)

$90,492

ΔQALY
+0.95 QALYs

ΔCOST
+$4,508

NET MONETARY BENEFIT
$90,492
ASSUMPTION-DRIVEN SCENARIO

UNCERTAINTY
not sampled (an alternative or cascade row)

DECISION
Return of the finding, then guideline management

COMPARATOR
The finding not returned

PERSPECTIVE
US, payer

POPULATION
Generally healthy adults given an incidental finding from sequencing for an unrelated indication

HORIZON
Lifetime

DISCOUNTING
3% a year

AGE
At the stated age of 30, assumed for this run, not observed; the source's value for identification at 45, held at 30

PRICE YEAR
Source 2013, restated to 2019 USD

VALUED PER
Carrier identified

Model basis: a published model's printed outputs, restated to 2019 US dollars with one cost adjustment that is not published here.

Key assumptions. At the stated age of 30, assumed for this run, not observed; the source's value for identification at 45, held at 30; penetrance not corrected for ascertainment; its source model was later corrected downward; age stated, not observed; age at identification.

Sources. PMID 25394171.

4 Alternative model: MSH2 carrier identified (man)

$32,766

ΔQALY
+0.3828 QALYs

ΔCOST
+$5,514

NET MONETARY BENEFIT
$32,766
ASSUMPTION-DRIVEN SCENARIO

UNCERTAINTY
not sampled (an alternative or cascade row)

DECISION
Colonoscopy every 2 years to 75

COMPARATOR
No surveillance

PERSPECTIVE
US, health-care sector (Medicare-based costs)

POPULATION
A man with an incidental MSH2 finding

HORIZON
Lifetime, to 80

DISCOUNTING
3% a year

AGE
At the stated age of 30, assumed for this run, not observed; if identified at 30

PRICE YEAR
Source 2019, restated to 2019 USD

VALUED PER
Carrier identified

Model basis: hypothetical: built from published components and stated assumptions, not a published model's result.

Genome-first screening value, a different question (not netted, not added): a Lynch-only test at 200 USD per person is not cost-effective in Guzauskas 2022 (ICER 153,200 USD per QALY without cascade testing), so its net monetary benefit is negative per person screened and per carrier found; the value above starts after sequencing, so the test cost is sunk.

Key assumptions. At the stated age of 30, assumed for this run, not observed; no surveillance: an assumption of this model, not the source's comparator arm; comparator: no surveillance; age stated, not observed.

Sources. PMID 35809086; PMID 33839100.

Where this document addresses each CHEERS 2022 item

CHEERS 2022 (Husereau et al., 2022) is a reporting standard for health economic evaluations. This maps its items to this dossier and the public repository. It records where each item is addressed, and which are not; it is not a claim of compliance. The model's inputs and structure are not published, so items that call for them are marked partly or not reported.

#	CHEERS 2022 ITEM	WHERE THIS DOCUMENT AND THE REPOSITORY ADDRESS IT	STATUS
1	**Title**	The document title names the analysis and the genome; it does not name the interventions compared	PARTLY
2	**Abstract**	The economic overview (page 2) of the public demonstration report is the structured summary	ADDRESSED
3	**Background and objectives**	The opening states the objective; each model card states the decision and its comparator	ADDRESSED
4	**Health economic analysis plan**	No analysis plan is published for these per-genome analyses	NOT REPORTED
5	**Study population**	Each model card: the source population, and the genome it is applied to	ADDRESSED
6	**Setting and location**	Each model card: the perspective and setting (mostly the United States)	ADDRESSED
7	**Comparators**	Each model card and the plane's table: the comparator of every decision	ADDRESSED
8	**Perspective**	Each model card	ADDRESSED
9	**Time horizon**	Each model card	ADDRESSED
10	**Discount rate**	Each model card: 3% a year unless the source states its own rate	ADDRESSED
11	**Selection of outcomes**	QALYs and costs, combined as net monetary benefit at $100,000 per QALY	ADDRESSED
12	**Measurement of outcomes**	Health effects are the published models' own or, for a model built here, computed from published components; none is measured in this genome	PARTLY
13	**Valuation of outcomes**	QALYs as each source, or the published components of a model built here, value them; no new preference elicitation	PARTLY
14	**Measurement and valuation of resources and costs**	Source costs, restated; resource use is the sources' own or, for a model built here, built from published components	PARTLY
15	**Currency, price date and conversion**	2019 US dollars; each card states the source's price year; costs are restated with a medical-care price index	ADDRESSED
16	**Rationale and description of model**	Each card names the method and the sources; the model's structure and inputs are not published	PARTLY
17	**Analytics and assumptions**	The key assumptions are stated on each card; the inputs and the full list of assumptions are not published	PARTLY
18	**Characterizing heterogeneity**	Per genome by construction; no subgroup or age-specific analysis is reported (the age is assumed)	NOT REPORTED
19	**Characterizing distributional effects**	Not reported	NOT REPORTED
20	**Characterizing uncertainty**	A 5th to 95th percentile on the counted value where the source prints ranges (parameter uncertainty only); alternative and cascade rows are not sampled	PARTLY
21	**Approach to engagement with patients and others affected by the study**	Not reported	NOT REPORTED
22	**Study parameters**	Not reported: the inputs and their values are not published; each card names its sources	NOT REPORTED
23	**Summary of main results**	The first page and the plane: every modeled value in its class	ADDRESSED
24	**Effect of uncertainty**	The percentile range on the counted value's card, where one is stored	PARTLY
25	**Effect of engagement with patients and others affected by the study**	Not reported	NOT REPORTED
26	**Study findings, limitations, generalizability and current knowledge**	The limitations page here; the public report's methods page; docs/LIMITATIONS.md	ADDRESSED
27	**Source of funding**	Not stated in this document	NOT REPORTED
28	**Conflicts of interest**	Not stated in this document	NOT REPORTED

Research scenario

HEOR TECHNICAL DOSSIER · HU4AE0F2

What the engine's models have been checked against

Three different things are kept apart: reproducing a source's own printed outputs, cross-checking against a model developed by others, and validating against observed outcomes. The last has not been done for any model here.

1 published table recomputed from its printed totals

1 model reproduces its source's printed outputs

1 model cross-checked against a model by others

0 models externally validated against observed outcomes

SECONDARY PUBLISHED-STRATEGY COMPARISON: COLONOSCOPY SCHEDULES AMONG KNOWN LYNCH CARRIERS

GENE AND THE SOURCE'S OPTIMAL STRATEGY (KASTRINOS 2021, TABLE 2)	NET MONETARY BENEFIT AGAINST THE NEXT EFFICIENT STRATEGY	NET MONETARY BENEFIT AGAINST THE LEAST INTENSIVE STRATEGY
MSH2 colonoscopy every 2 years from age 25	**1,892 USD** **+0.028 QALYs, +908 USD**	**91,531 USD** **+0.664 QALYs, −25,131 USD**

Kastrinos 2021 compares colonoscopy schedules among carriers who are already known; it prints no 'not identified' arm, so it is not the value of identifying a carrier and is carried beside each Lynch decision, never as its value. The same optimal strategy has two net monetary benefits, because it is set against two comparators. For MLH1 they are 4,030 USD against the next efficient strategy (the source's own incremental comparison) and 225,191 USD against the least intensive strategy listed (current management for a family history of colorectal cancer, which the source rates dominated). Neither is the other's estimate. All 20 printed strategies are classified as the source classifies them and each gene's optimal strategy is the highest net monetary benefit at the source's threshold; 3 of the 4 printed ICERs reproduce from the printed totals, and 1 does not (see the reproduction kit in the repository). Discounting is unresolved: the source states 3% a year, but its printed QALYs exceed the most that 3% discounting can give over ages 25 to 75 (about 26); every figure here is arithmetic on the totals as printed.

WHAT IS NOT SHOWN HERE

Two further checks are described in `docs/VALIDATION.md`, but their figures are not published in this edition: the comparison of the engine's own model with this source, strategy by strategy, and the quantity-by-quantity comparison of a rebuilt hemochromatosis model with its source's printed untreated-cohort outputs. Their figures would show the model's outputs beyond what the demonstration reports print.

WHICH VALUES REST ON A PUBLISHED MODEL

27 registered values (rows, not independent studies) rest on 6 published analyses, and 71 are hypothetical, assumption-based scenarios built here from published components or stated assumptions (60 of them first-pass pediatric models), not a published model's result. None is externally validated. The drug-response decisions are assembled from published inputs and carry no basis label.

Not established. No model here has been validated against observed outcomes in a population. A cross-check shows how two models agree, not which is right; a reproduction shows the arithmetic matches the source, not that the source's model predicts well.

What was verified, and what was not

Verified for this document

- ✓ Every dollar figure and QALY figure here is a value the engine stored for this genome, or is calculated here from stored terms and printed with them (a cost per QALY gained).
- ✓ Net monetary benefit equals the health gain valued at the threshold, less the change in cost, on the stored terms (within 5 USD; the printed terms are rounded).
- ✓ This document is made by a separate export that prints a fixed list of fields. Its check is that every decimal, and every whole number of 100 or more apart from years and the plane's axis marks, also appears in this genome's public demonstration reports, the public documents or the reproduction kit's recorded output; the engine's own consistency checks are not extended to it.
- ✓ Every page is checked for overflow and clipping.

Not established

- External validation: none of the decision models has been validated against observed outcomes in a population.
- Clinical validity of the calls: research-genome calls, not confirmed by a clinical laboratory.
- Donor applicability: the age 30 is stated for this run, not observed (it is the age at which the published source reports its result), not the donor's.
- Per-decision probabilistic analysis: a range is shown for the counted value where its source prints ranges; alternative and cascade rows are not sampled, and an interval printed is parameter uncertainty only.

LIMITATIONS

- Each value is either a published model's printed outputs, restated, or a hypothetical scenario built here from published components; the card says which. The assumptions are the card's, and a source population may differ from a genome-first finding.
- Values in different classes, to different beneficiaries, are never added; no total across this dossier is meaningful.
- The model's inputs, structure and the registry behind it are not published; this edition does not describe a method sufficient to reproduce the analysis.

Proprietary. Demonstration report for a public research genome. GenomeLens, its economic models and its methods are proprietary. This report shows results, not the method: it is not open-source software, not a description sufficient to reproduce the analysis, and not a license to use either.